

Incomplete, continuous, non-Archimedean

Model (conjectured) — ai; produced by Claude Fable 5.1 (Cowork session).

0.1 Package

- **Transitivity.** For all $p, q, r \in \Delta(X)$: if $p \geq q$ and $q \geq r$ then $p \geq r$.
- **Mixture continuity.** For all $p, q, r \in \Delta(X)$ the sets $\{\lambda \in [0,1] : \lambda p + (1-\lambda)q \geq r\}$ and $\{\lambda \in [0,1] : r \geq \lambda p + (1-\lambda)q\}$ are closed.
- **$\neg$ Archimedean.** For all $p, q, r \in \Delta(X)$: if $p > q > r$ then there are $\alpha, \beta \in (0,1)$ with $\alpha p + (1-\alpha)r > q$ and $q > \beta p + (1-\beta)r$.

0.2 Notes

Completeness is used essentially in continuity-completeness-imply-archimedean; no countermodel yet.

Record: topics/decision-theory/models/incomplete-continuous-non-archimedean.yaml